Procedural Spider Web System

User Manual for Unity Scene Setup

Procedural Spider Web System creates spider webs between attachment points in a Unity scene.

There are two ways to use it:

- Place WebAnchor objects by hand for an art-directed web.
- Leave the anchor list empty and let the generator find nearby collider surfaces with raycasts.

For final scenes, manual anchors are recommended. They give better control over the web outline, attachment points, sagging, and decorative droplets.

Package Contents

Script

- ProceduralSpiderWeb.cs - the main web generator.
- WebAnchor.cs - an attachment point used by the generator.
- SpiderWebDemoAnimator.cs - a small demo script for subtle object motion.

Prefab

- ProceduralSpiderWeb.prefab - ready-to-use generator object.
- Anchor.prefab - simple attachment point.
- WebAnchorHook.prefab - decorative metal hook with an anchor setup.
- SilkDroplet.prefab - small droplet detail for web strands.

Materials

- M_SpiderSilk.mat - material for web strands.
- M_DewGlass.mat - material for droplets.
- M_AgedWood.mat, M_CoolStone.mat, M_DarkIron.mat - demo environment materials.

Model

- OBJ models for the hook, droplet, aged beam, broken board, and stone ledge.

Scenes

- ProceduralSpiderWeb_Demo.unity - a ready demo scene showing the intended setup.

Quick Start

1. Open the scene where you want to add a spider web.
2. Drag Prefab/ProceduralSpiderWeb.prefab into the Hierarchy.
3. Drag several Prefab/Anchor.prefab or Prefab/WebAnchorHook.prefab objects to the places where the web should attach.
4. Select the ProceduralSpiderWeb object.
5. In Explicit Anchors, set the list size and drag all anchor objects into the list.
6. Assign Materials/M_SpiderSilk.mat to Silk Material.
7. Assign Prefab/SilkDroplet.prefab to Droplet Prefab if droplets are needed.

8. Open the component context menu and choose Generate Spider Web.

If Live Preview In Editor is enabled, the web updates automatically when settings are changed.

Placing Anchors

An anchor is the point where the web is attached to the environment.

Place anchors around the outer edge of the desired web shape:

- on wooden beams;
- on stones;
- on walls;
- on metal hooks;
- on window corners, cave edges, door frames, or old props.

At least 3 anchors are required. A natural web usually looks best with 8 to 14 anchors placed unevenly. Avoid a perfect circle unless the scene specifically needs a stylized web.

Good starting layout:

1. One anchor at the upper left.
2. One anchor near the upper center.
3. One anchor at the upper right.
4. Two anchors on the left side at different heights.
5. Two anchors on the right side.
6. Two or three anchors along the bottom edge.

After placing them, add the objects to the Explicit Anchors list on the generator.

ProceduralSpiderWeb Settings

Explicit Anchors

Manual attachment points. This is the main workflow for finished scenes.

If this list has objects assigned, the generator uses only those anchors.

Surface Mask

Layer mask used for automatic surface search. This is only used when Explicit Anchors is empty.

For raycast mode:

1. Add colliders to walls, beams, or props.
2. Make sure those objects are on a layer included in Surface Mask.
3. Place the generator near the center of the intended web area.
4. Use Generate Spider Web.

Max Anchor Distance

Maximum search distance for raycast mode. Increase this value if the generator cannot find nearby surfaces.

Max Anchors

Limits how many attachment points are used. For manual setups, 8 to 16 usually works well.

Raycast Samples

Number of directions used for automatic surface search. Higher values can find more complex surrounding geometry, but generation becomes slightly more expensive.

Ring Count

Number of circular strands inside the web.

- 3 - 5 - simple web.
- 6 - 9 - detailed normal web.
- 10+ - dense web for larger props.

Sag

Downward sag applied to the strands.

- 0.05 - 0.15 - fresh, tighter web.
- 0.18 - 0.30 - natural old web.
- 0.35+ - heavy sag for abandoned areas.

Irregularity

Shape unevenness. Keep this moderate for realistic results.

Recommended range: 0.08 - 0.18.

Seed

Controls deterministic randomness. Change the seed when the current shape is close but you want a different variation.

Silk Material

Material used by the generated LineRenderers. Use M_SpiderSilk.mat or a custom transparent material.

Line Width

Strand thickness in world units. For a normal Unity scale, 0.008 - 0.016 works well.

Droplet Prefab

Prefab spawned on selected strands as dew or small detail.

You can use:

- SilkDroplet.prefab;
- small dust or dew meshes;
- a custom transparent mesh.

Droplet Count

Maximum number of droplets. Use fewer droplets for dark scenes or distant webs.

Setting Up a Web in Your Scene

1. Build the surrounding environment: wall, beams, rocks, props, or a window frame.
2. Make sure objects use a reasonable Unity scale. Very small or very large scale makes strand width harder to tune.
3. Add Anchor.prefab to the attachment points.

4. Use WebAnchorHook.prefab where a visible hook is useful.
5. Add ProceduralSpiderWeb.prefab near the center of the future web.
6. Fill the Explicit Anchors list.
7. Assign M_SpiderSilk.mat to Silk Material.
8. Assign SilkDroplet.prefab to Droplet Prefab if needed.
9. Run Generate Spider Web.
10. Adjust Ring Count, Sag, Irregularity, and Line Width.
11. Keep Generate On Start enabled for runtime generation, or disable it if the web is already generated and should remain static.

Generate On Start

Enable it when:

- anchors move during gameplay;
- the scene is built at runtime;
- the web should be rebuilt when the scene starts.

Disable it when:

- the web is static;
- the web has already been generated in the editor;
- no runtime regeneration is needed.

Live Preview In Editor

Enable it while tuning the web shape.

Disable it when:

- the scene is large;
- many webs are present;
- setup is finished and automatic editor rebuilds are no longer needed.

Troubleshooting

The web does not appear

Check that:

- Explicit Anchors contains at least 3 anchors;
- assigned anchors still exist in the scene;
- Silk Material is assigned;
- Line Width is not too small;
- the generator object is active.

The web looks too perfect

Increase Irregularity to 0.12 - 0.18 and move a few anchors slightly.

The web sags too much

Decrease Sag.

Droplets are not visible

Check Droplet Prefab, material transparency, prefab scale, and Droplet Count.

Raycast mode does not find surfaces

Check that:

- surfaces have colliders;
- surfaces are on a layer included in Surface Mask;
- Max Anchor Distance is high enough;
- the generator is placed between the surfaces, not far away from them.

Practical Tips

- Manual anchor placement gives the best final result.
- Do not place anchors in perfect symmetry.
- For large webs, increase Ring Count and tune Line Width.
- For horror scenes, use more Sag and fewer droplets.
- For a wet morning look, use thin strands and more droplets.
- If the web is placed against a bright wall, make the silk material slightly less transparent or a bit darker.

Recommended Starting Values

Small web:

- Ring Count: 5
- Sag: 0.12
- Irregularity: 0.10
- Line Width: 0.008
- Droplet Count: 6

Medium web:

- Ring Count: 7
- Sag: 0.18
- Irregularity: 0.13
- Line Width: 0.011
- Droplet Count: 14

Large old web:

- Ring Count: 9
- Sag: 0.26
- Irregularity: 0.18
- Line Width: 0.014
- Droplet Count: 20

Demo Scene

Open Scenes/ProceduralSpiderWeb_Demo.unity to see a complete setup.

The demo scene includes:

- environment objects;
- anchors;
- configured generator;
- silk material;
- droplet prefab;
- camera and lighting.

The fastest way to understand the setup is to open the demo scene, select ProceduralSpiderWeb, and inspect the anchors assigned in the Explicit Anchors list.